ECE264D:
CMOS Analog Integrated Circuits and Systems IV
PLL Architecture and Design

Lecture 1:
Introduction

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ECE264D Course Information

- Goals of the course
  - Acquire the fundamentals of frequency synthesizer design and analysis
  - Get familiar with the state of the art, ready to do synthesizer related research
  - Understand challenges of synthesizer design in commercial transceivers, ready to ramp up and work in a design team
  - Prerequisite: analog IC design, signal processing fundamentals.
- Design tools
  - Matlab for modeling and system level design/simulation
  - Cadence for transistor level design/simulation
- Topics of the course
  - Fundamentals covered in classical textbooks
  - Practical issues important to both industry design engineers and graduate researchers

References

- Recommended textbooks
  - B. Razavi, Design of CMOS Phase-Locked Loops: From Circuit Level to Architecture Level, 2020
- References:
  - R. B. Staszewski, All-Digital Frequency Synthesizer in Deep-Submicron CMOS, 2006.
  - W. Egan, Advanced Frequency Synthesis by Phase Lock, 2011
  - W. Rhee, Phase-Locked Frequency Generation and Clocking, 2020
  - Other references, including books and papers, will be listed at the end of the notes of each lecture. The PowerPoint slides and assigned IEEE papers will play the role of a textbook for ECE264D.

Course Syllabus

- Phase locked loop overview
  - PLL for frequency synthesizers
  - PLL for clock generations
  - PLL for CDR in wireline communications
  - Design challenges
- Analog phase locked loop
  - Building blocks
    - Phase detector (PD), Phase frequency detector (PFD)
    - Charge pump (CP)
    - Loop filter
    - Voltage controlled oscillator (VCO)
    - Frequency divider
  - S-domain model, stability analysis, noise transfer functions
  - VCO design and phase noise analysis

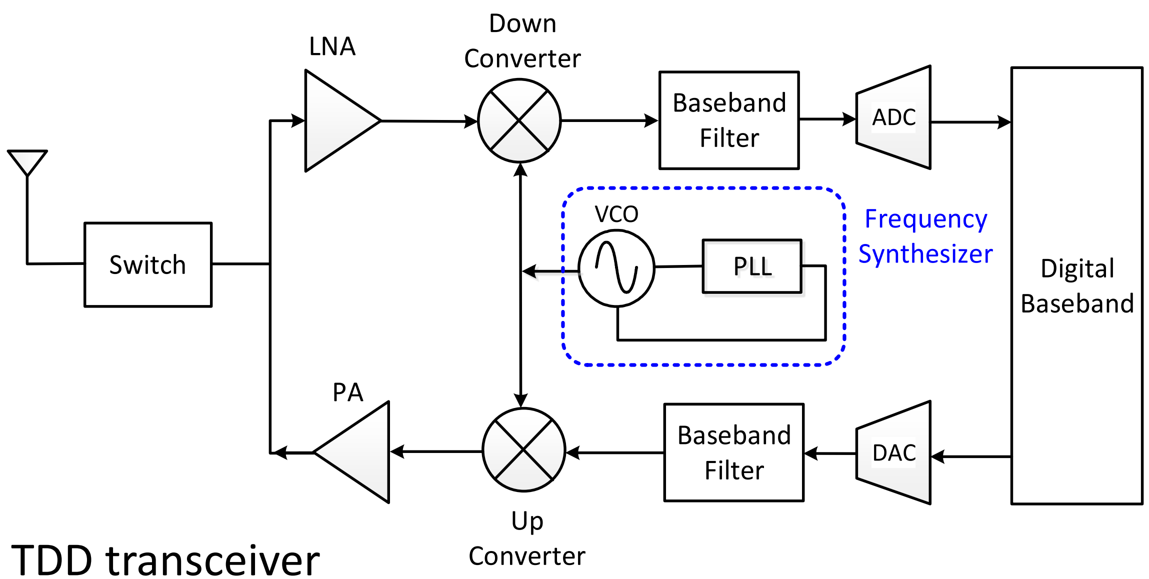
Course Syllabus

- Clock generation and frequency synthesis
  - Integer and fractional PLL
  - Delta Sigma Modulator (DSM) and noise shaping
- All digital PLL
  - Building blocks
    - Time to digital converter (TDC), phase to digital converter (PDC)
    - Digital loop filter
    - Digital controlled oscillator (DCO)
  - Quantization noise analysis
  - Z-domain model and analysis
- Delay locked loops (DLL)
  - DLL modeling
  - Multiplying DLL

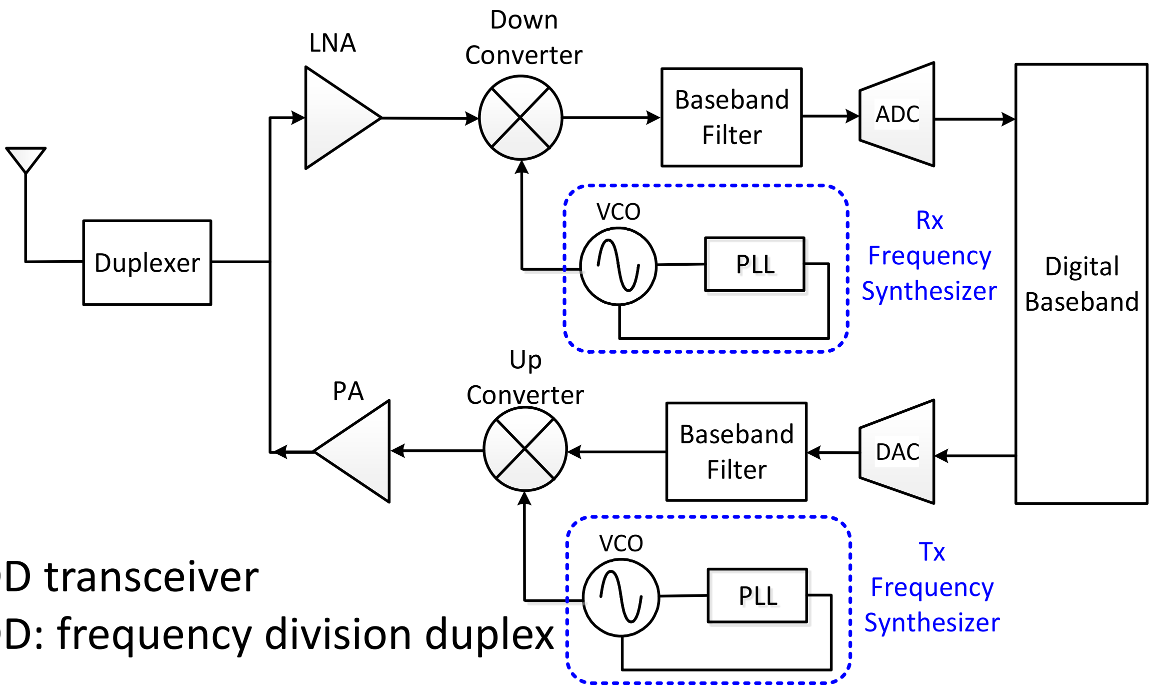
Course Syllabus

- Millimeter wave frequency generation
  - Challenges in millimeter wave frequency oscillator and divider design
  - Frequency multipliers
  - Phase shifters/rotators
  - FMCW (frequency modulated continuous wave) radar

PLL as frequency synthesizers in wireless transceivers



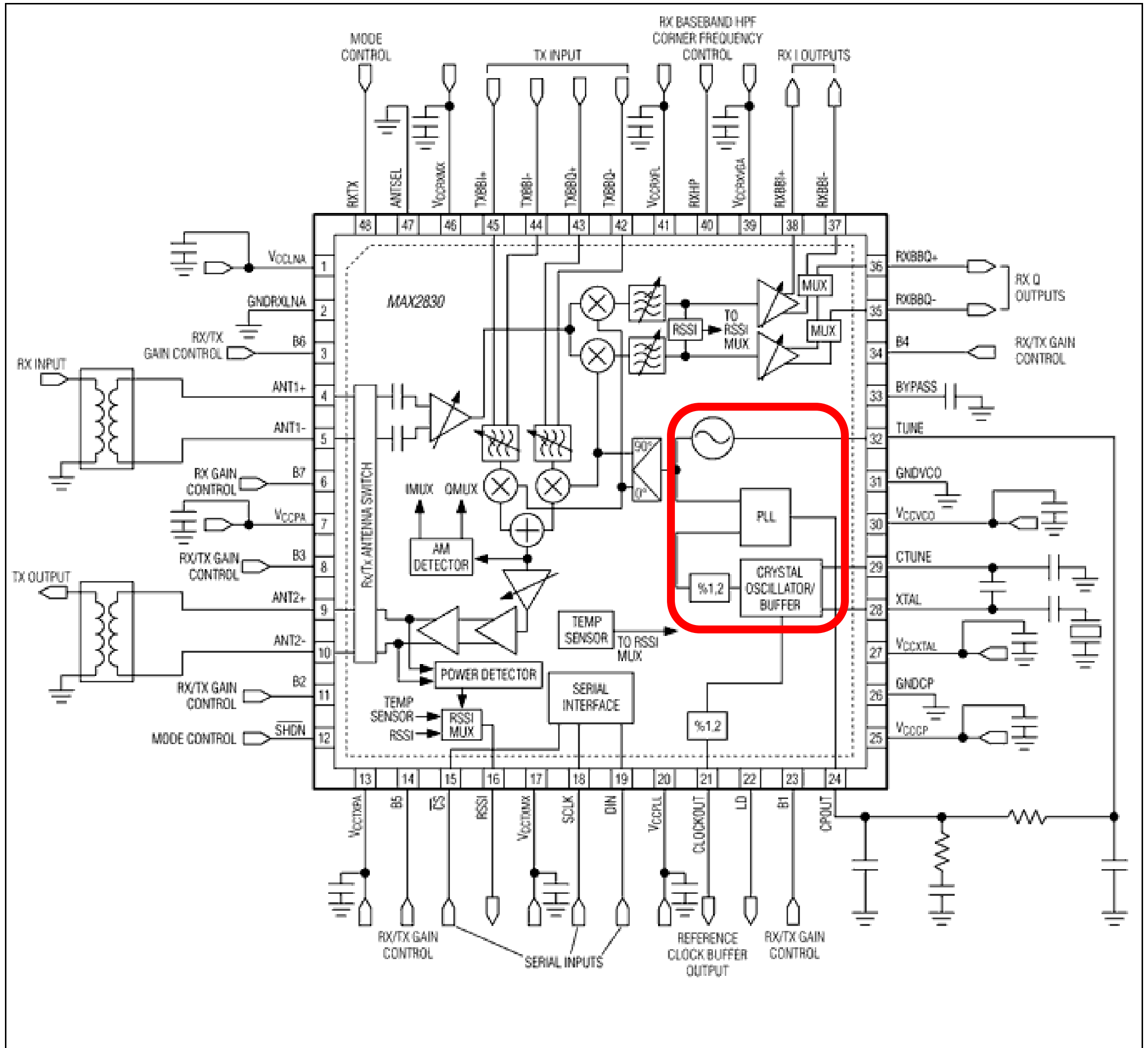
TDD transceiver
TDD: time division duplex



FDD transceiver
FDD: frequency division duplex

- Frequency synthesizer generates local oscillation (LO) signal that drives Rx down conversion mixer or Tx up conversion mixer, or both.
- In TDD transceiver, one synthesizer can be shared between Rx and Tx signal paths
- In FDD transceiver, each signal path needs its own synthesizer at different LO frequencies.

PLL as frequency synthesizer in wireless transceivers



NOTE: ALL GROUND PINS (PINS 2, 26, AND 31) AND BYPASS CAPACITORS' GROUND REQUIRE THEIR OWN VIAS TO GROUND. DO NOT CONNECT THEM TO THE EXPOSED PADDLE GROUND.

- Synthesizer generates local oscillation (LO) signal that drives down-converter and up-converter
- Single synthesizer in **TDD** transceiver: one synthesizer shared between receiver and transmitter
- Off-chip crystal and on-chip crystal oscillator provides highly accurate reference signal for frequency synthesis
- On-chip VCO is locked by phase locked loop (PLL) at integer or fractional multiple of reference frequency

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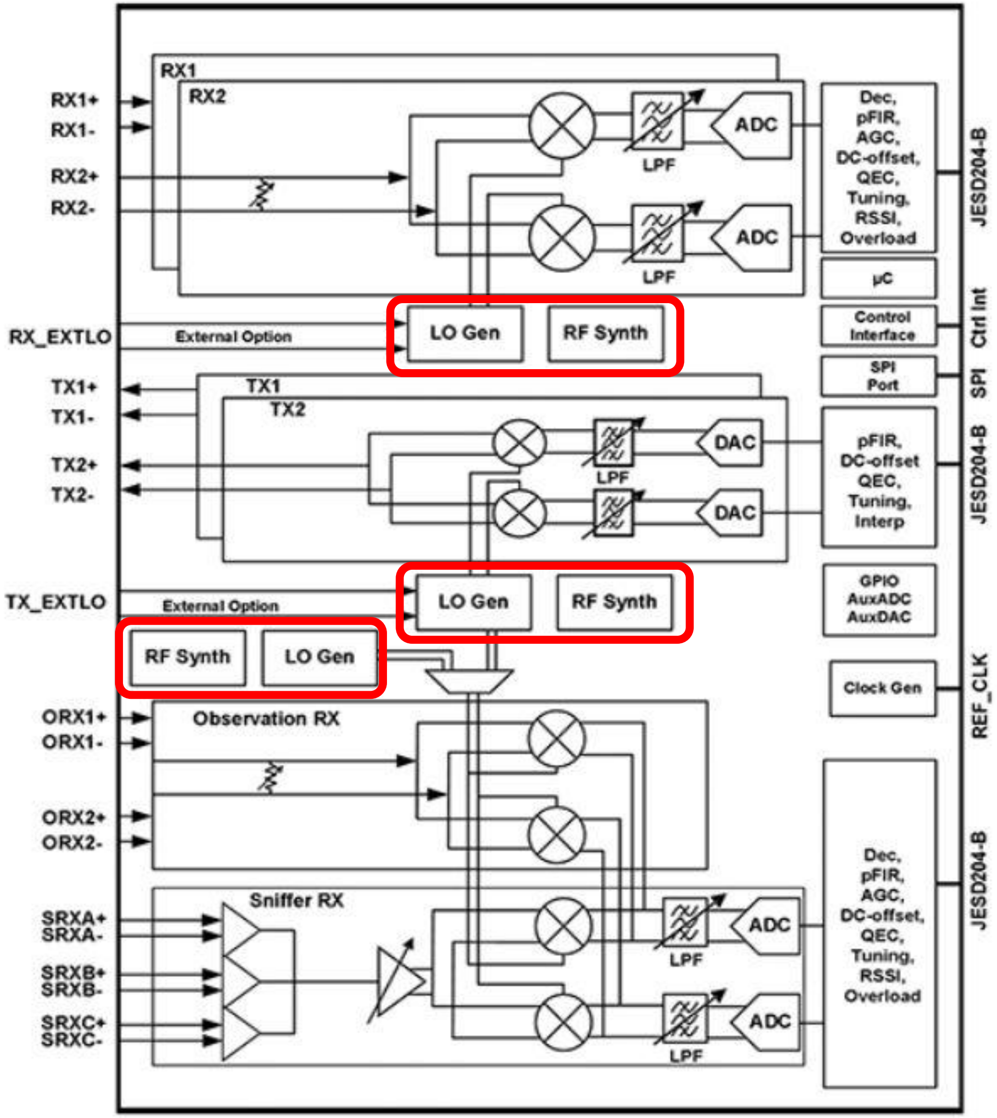
Ref: Maxim MAX2830 2.4GHz 802.11g/b RF transceiver (WLAN)

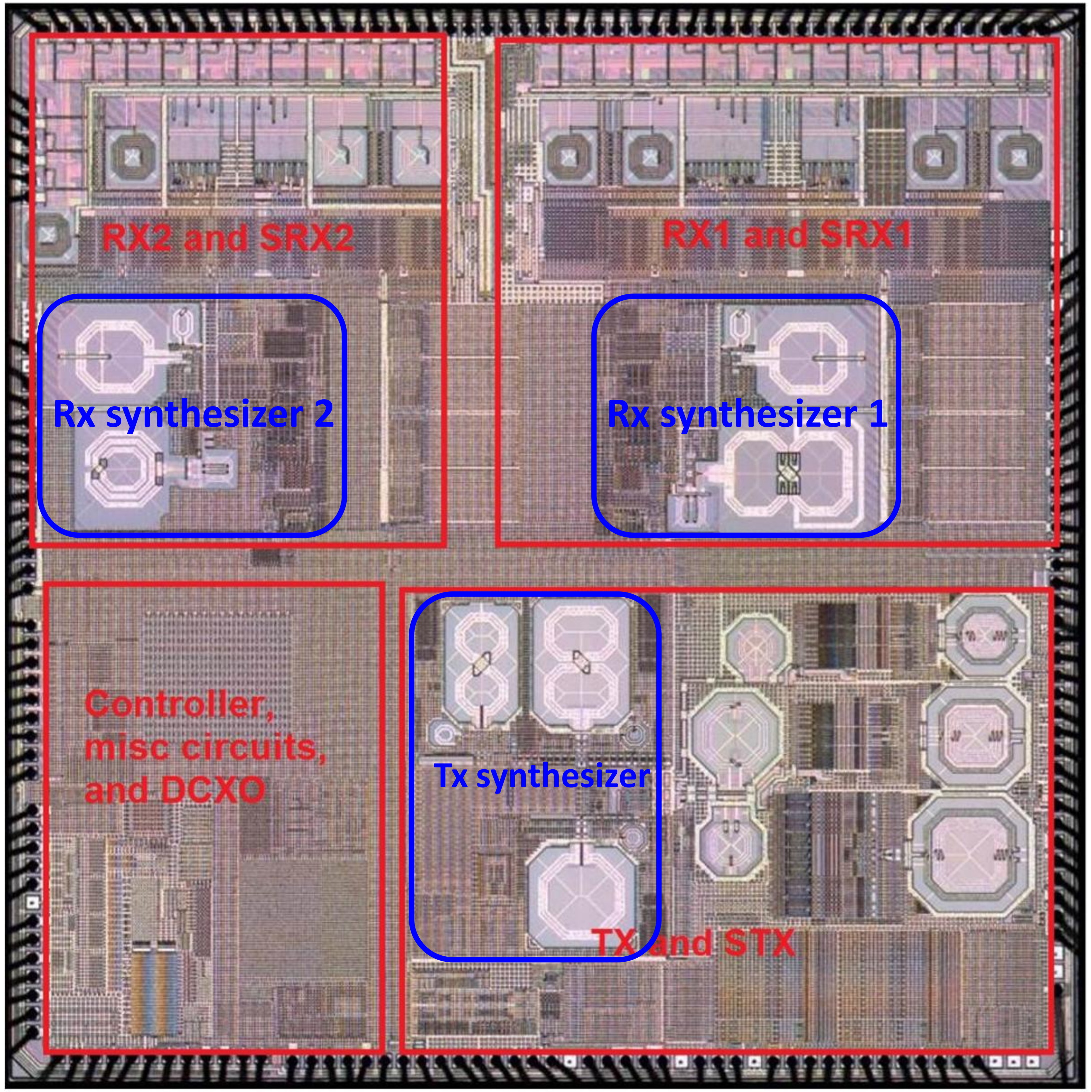
PLL as frequency synthesizer in wireless transceivers



- Separate Rx and Tx synthesizers in **FDD** transceiver
- Multiple synthesizers for multiple receivers or transmitters if they operate simultaneously and at different frequencies

Ref: Microwave Journal, AD9371 ISM band transceiver

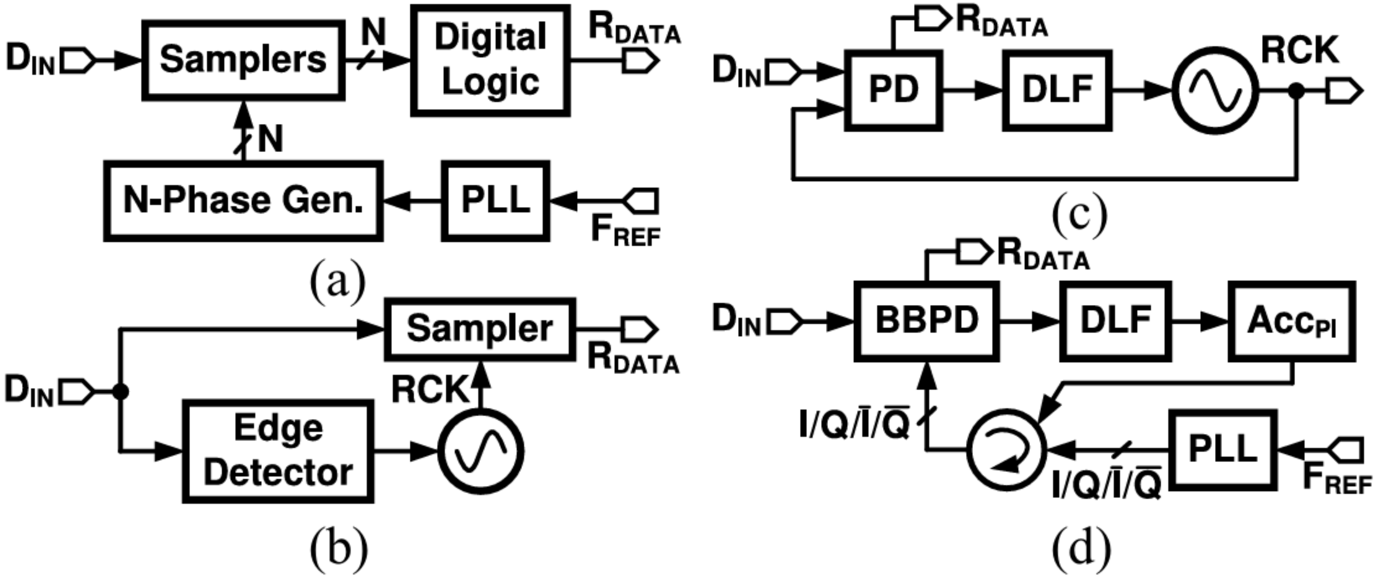
PLL as frequency synthesizer in wireless transceivers



[Ref 6]: MediaTek, ISSCC 2017

- A 40nm Low-Power Transceiver for LTE-A Carrier Aggregation
- Two Rx synthesizers with two VCOs each and one Tx synthesizer with three VCOs
- To implement carrier aggregation in 4th and 5th generations (4G, 5G) cellular communications, it is very common to have 3~10 synthesizers integrated on a single chip. The number is still increasing.
- Challenges: VCO to VCO pulling, VCO to signal paths coupling, LO routings, etc.

PLL as clock and data recovery in wireline receivers



[Ref 8]

PLL is used in various types of conventional CDRs

(a) blind oversampling

(b) edge detector and gated VCO

(c) PLL-based clock recovery

(d) PI-based clock recovery

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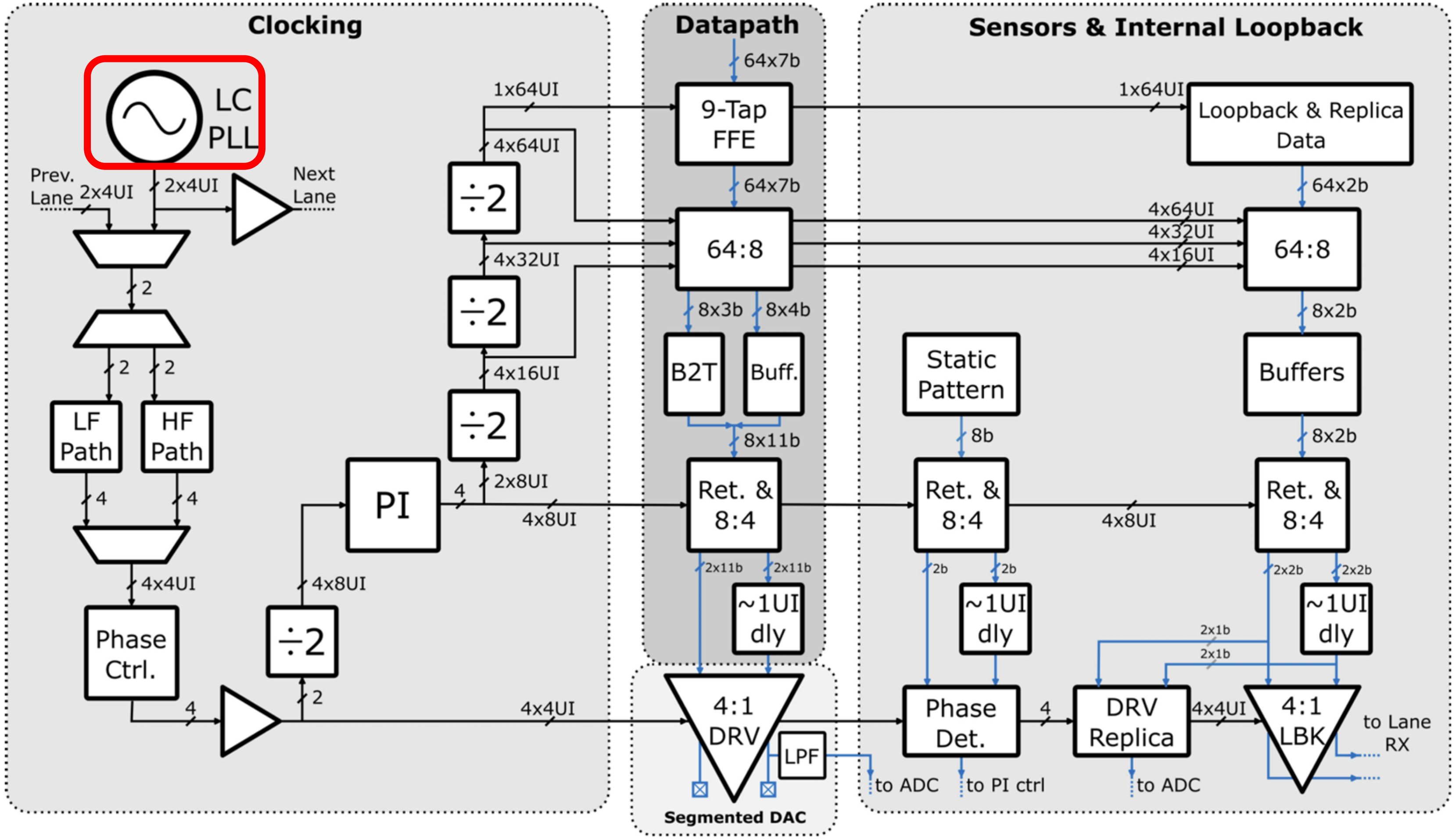
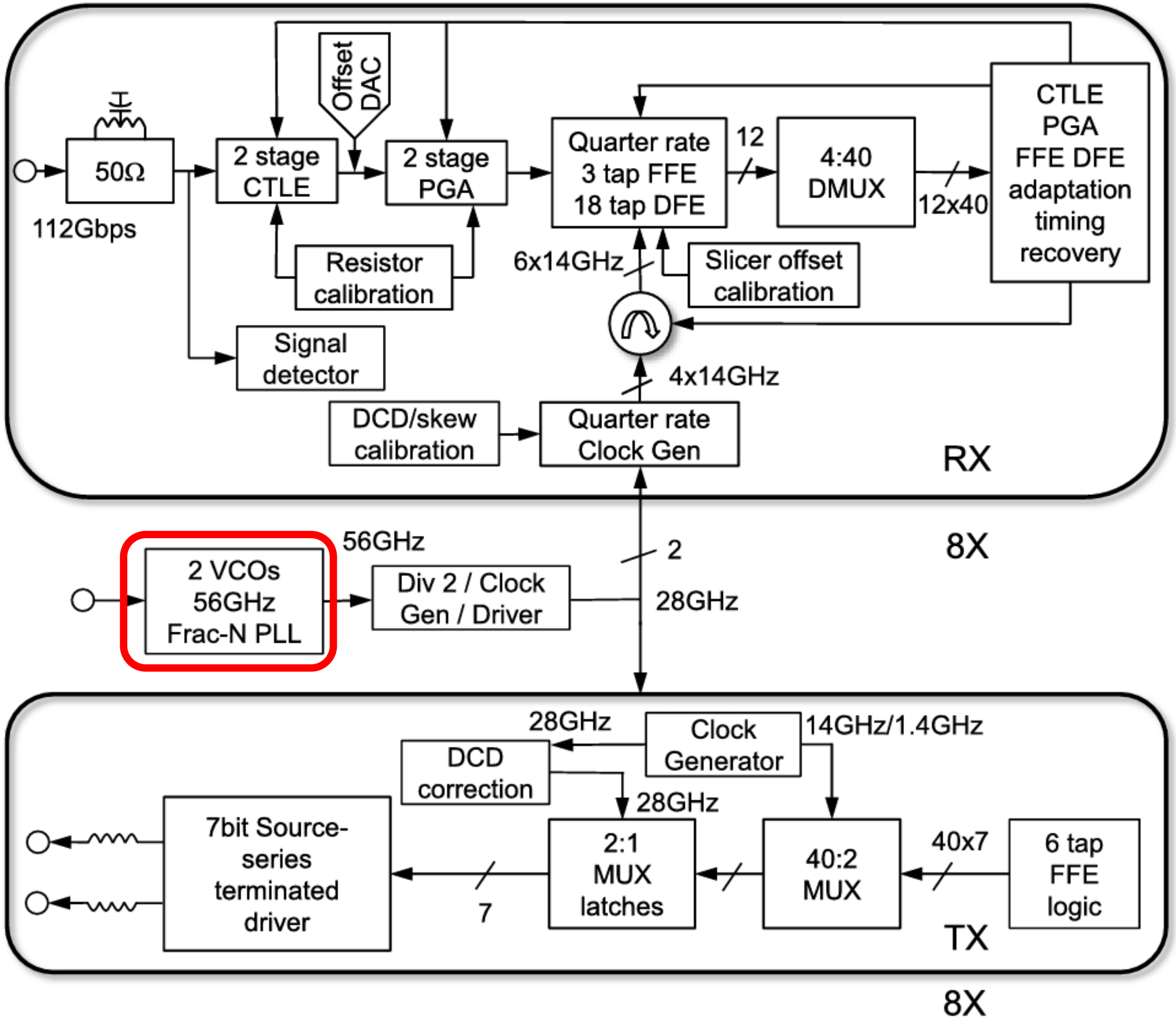
PLL as clock generation in wireline transmitters



[Ref 9]

- A digital LC PLL as clock generation
- PLL produces a four unit-interval (UI) differential clock driving the two quadrature clock generation blocks used to support different data rates
- Multiphase generation and clock distribution networks

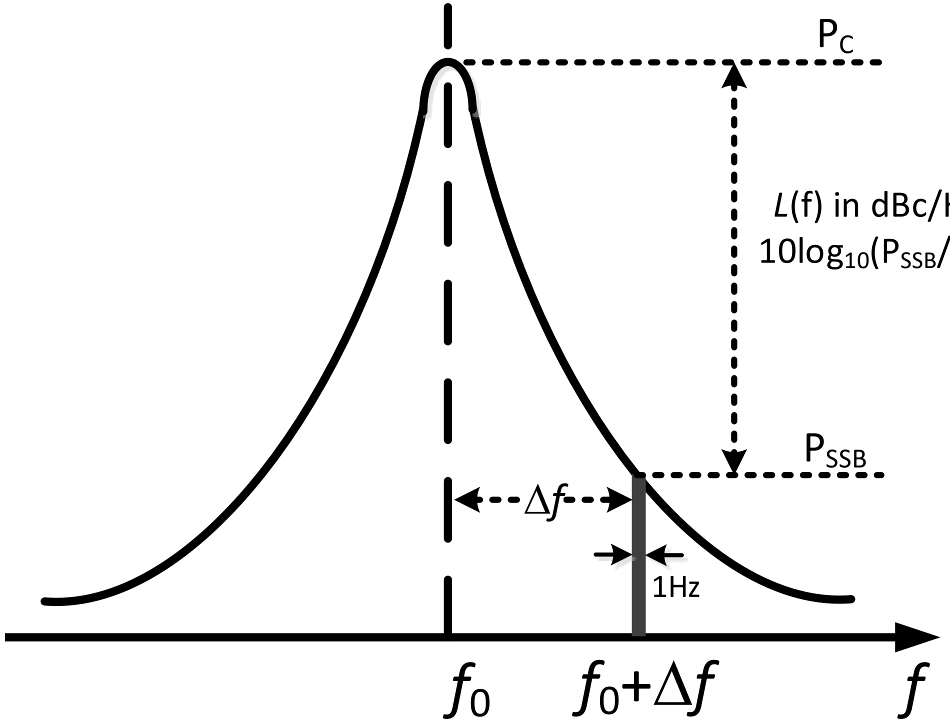
PLL as clock generation in wireline transceivers



- Fractional-N PLL as clock generation for both receiver and transmitter
- DSP based clock and data recovery
- Clock divider, quadrature phase generation and clock distribution networks

[Ref 10]

PLL design specifications: *Phase noise*



<u>Phase noise (PN) definition</u>

The ratio between the noise power in unit bandwidth at a given frequency offset and the carrier power in decibel

The unit of phase noise is dBc/Hz

To clearly specify phase noise, we need to know

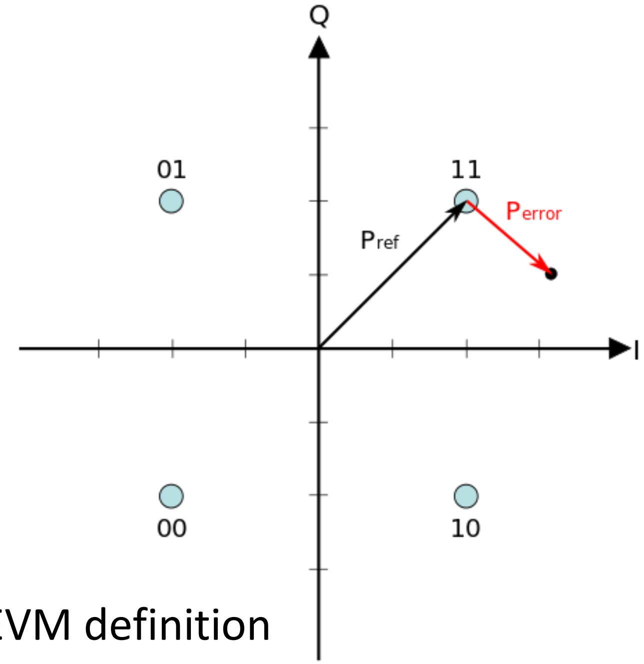
- Carrier power (phase noise in dBc, 'c' is carrier)
- Offset frequency
- Single sideband (SSB) or double sideband (DSB)
  - Use SSB for out of band spot phase noise and DSB for in band integrated phase noise (IPN). Why?

In LNA case, noise factor is defined as

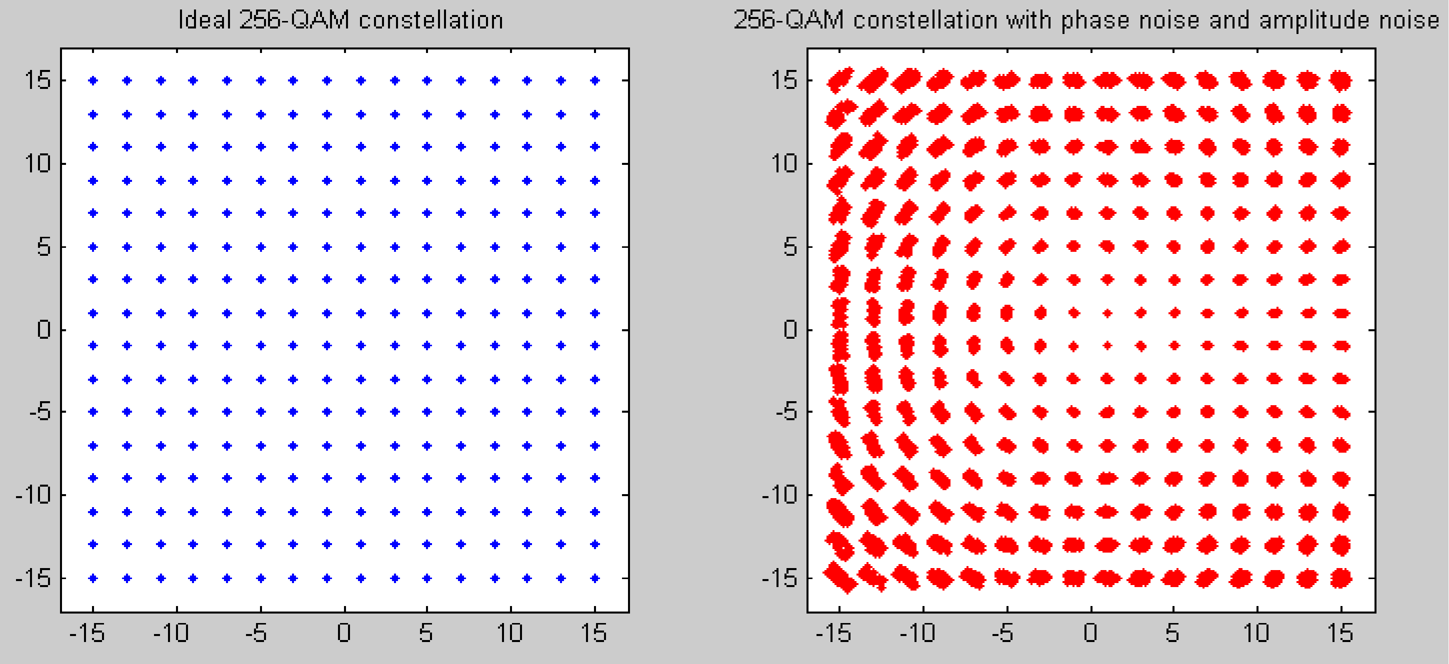
$$F = \frac{\mathrm{SNR_i}}{\mathrm{SNR_o}} = \frac{\frac{S_i}{N_i}}{\frac{S_o}{N_o}} \qquad \frac{S_o}{N_o} = \frac{S_i G}{N_a + N_i G} \qquad F = 1 + \frac{N_a}{N_i G}$$

SNR, signal to noise ratio, is a relative quantity

PLL design specifications: *Integrated phase noise*



EVM definition

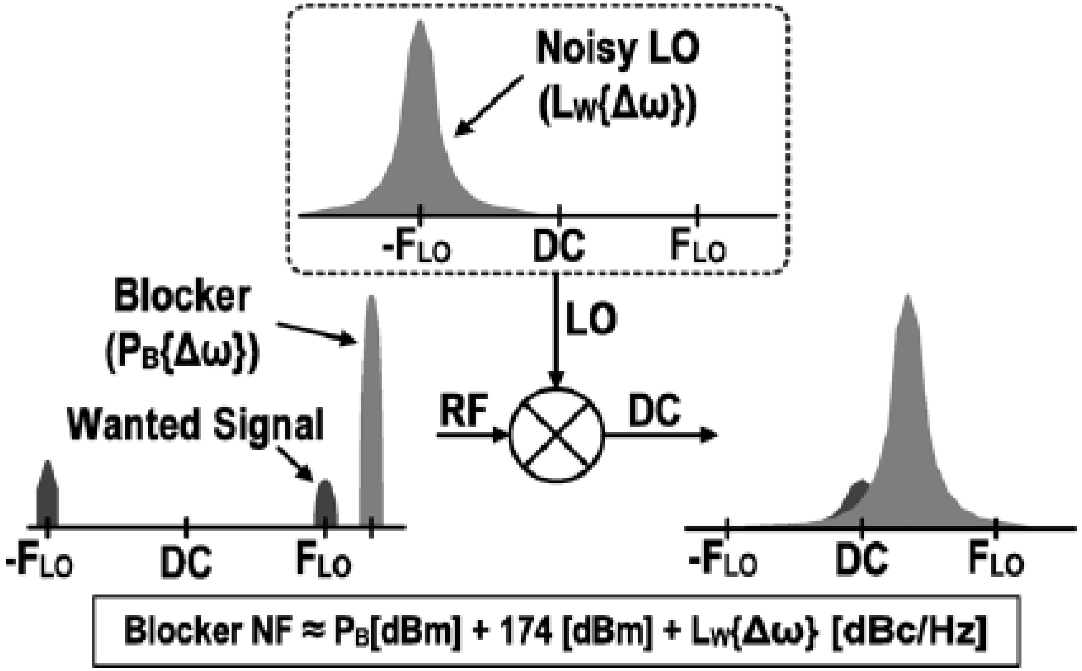


- In band integrated phase noise (IPN) is the integration of phase noise in signal bandwidth
- IPN is driven by signal modulation Error vector magnitude: $EVM(dB) = 10log_{10}\left(\frac{P_{error}}{P_{ref}}\right)$
- IPN degrades EVM and brings adjacent constellation points closer that causes demodulation failure when below certain threshold.
- Lower IPN is required when higher modulation scheme is used (QPSK, 16QAM, 64QAM, 256QAM, 1024QAM)

PLL design specifications:

Spot phase noise

- Spot phase noise (SPN) is the phase noise at a given offset frequency
- Factors that drive SPN requirement
  - Receiver LO
    - Reciprocal mixing: adjacent channel jammers mixing with LO phase noise to generate in band noise and degrade SNI Spec depends on jammer location and strength
    - In FDD transceiver, the largest jammer is its own transmitter signal located at duplex frequency offset
  - Transmitter LO
    - Phase noise at duplex frequency offset directly leaks to receiver input due to finite isolation in FDD transceiver
    - Stringent emission mask for transmit signal at antenna to limit jammer level to other receivers



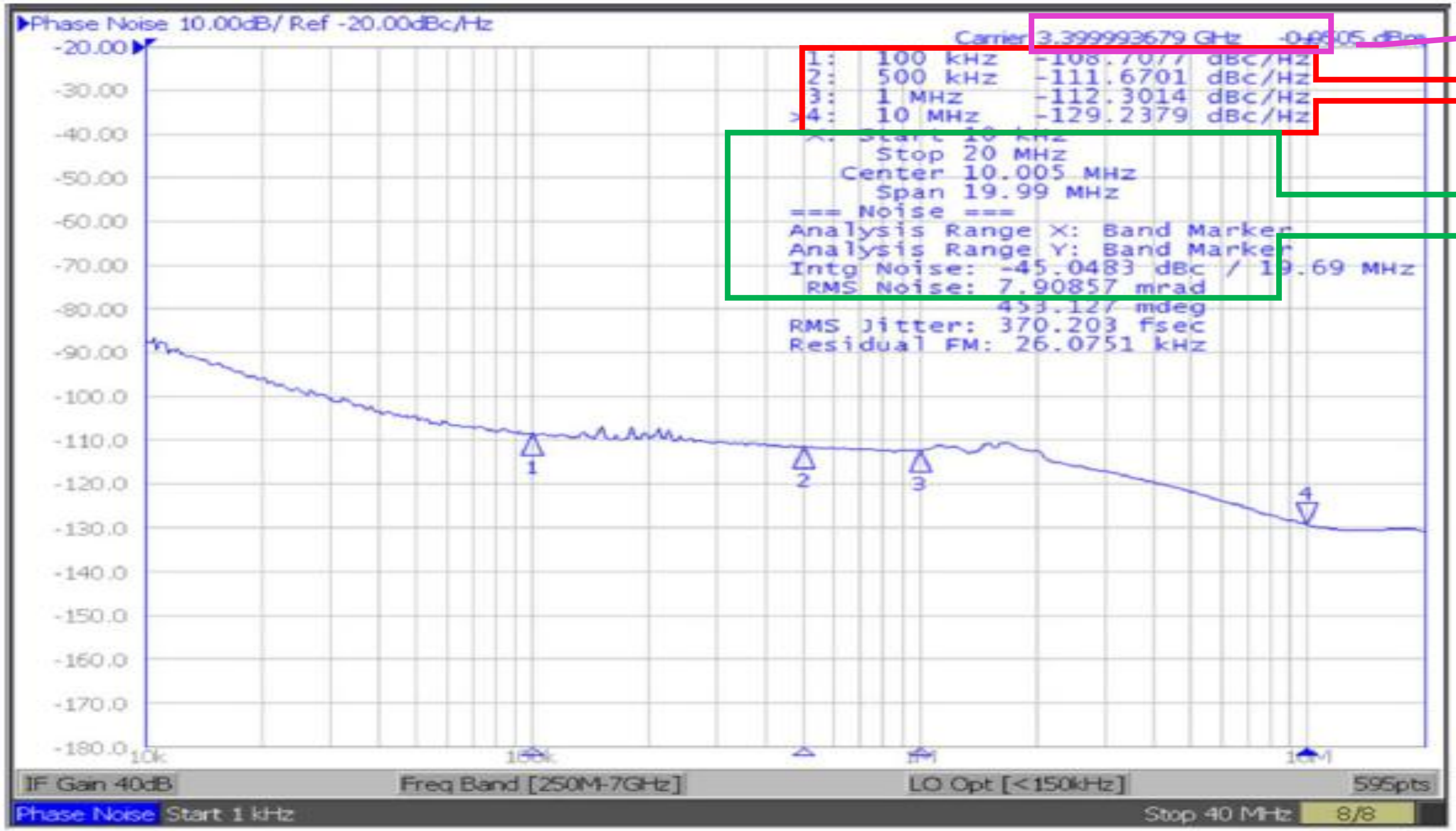
Reciprocal mixing mechanism

Ref [4]

PLL design specifications:

Phase noise

A typical phase noise profile



Carrier frequency: 3.399GHz

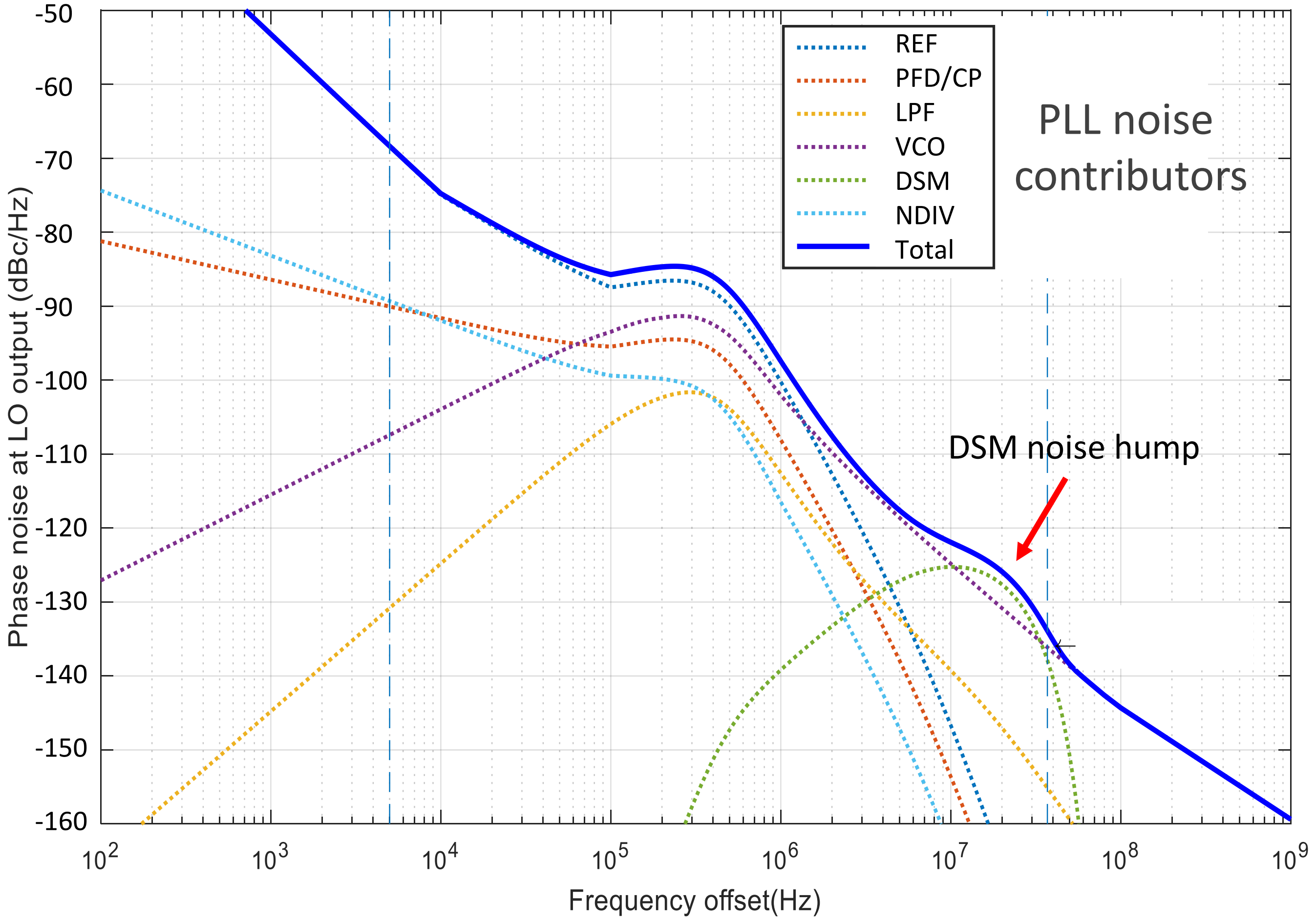
Spot phase noise in dBc/Hz at 100k, 500k 1M, 10MHz

Integrated phase noise in dB
Integration frequency range 10k ~ 20MHz

Ref [5]

PLL design specifications:

How to partition PN spec budget



- Main phase noise contributors
  - Input reference noise (REF)
  - PLL building block noise (PFD/CP, LPF, DSM, DIV)
  - VCO noise
- Reference path noise low pass filtered
- VCO noise high pass filtered
- DSM noise dominates at band edge
- Optimal loop bandwidth balances all contributors

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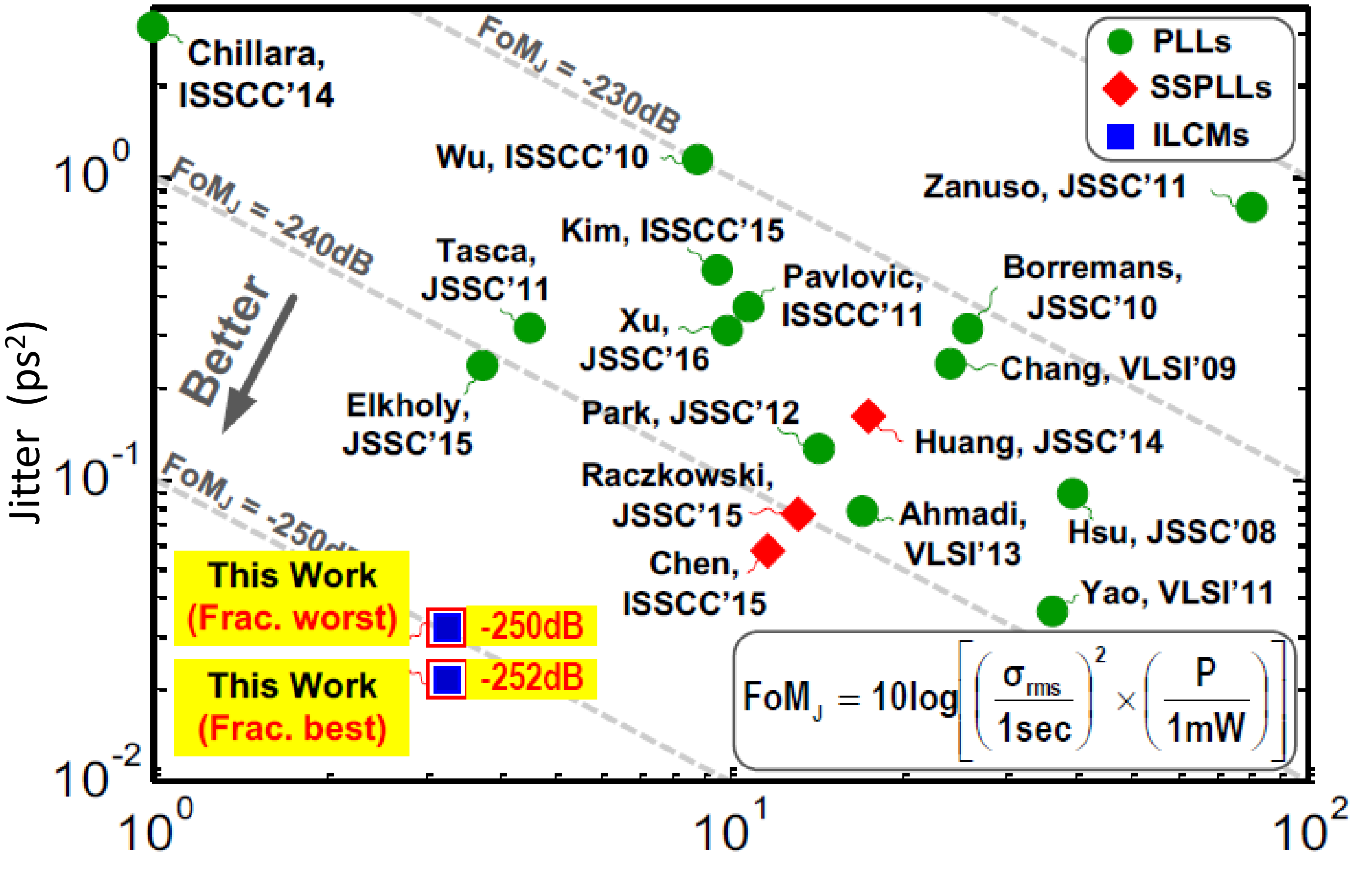
PLL design figure of merit



- Power consumption and die area are not part of specs but important for all practical applications
- Figure of merit (FOM) for comparison across synthesizers with different requirements
- FOM

$$= 10log_{10}\left[\left(\frac{\sigma_{rms}}{1\ sec}\right)^2 \cdot \left(\frac{P}{1\ mW}\right)\right]$$

- FOM_A

$$= 10log_{10}\left[\left(\frac{\sigma_{rms}}{1\ sec}\right)^2 \cdot \left(\frac{P}{1\ mW}\right) \cdot \left(\frac{A}{1\ mm^2}\right)\right]$$

- Jitter σ_{rms} is related to IPN by

$$\sigma_{rms} = \frac{\sqrt{IPN}}{2\pi \cdot f_{VCO}}$$

Ref [3]

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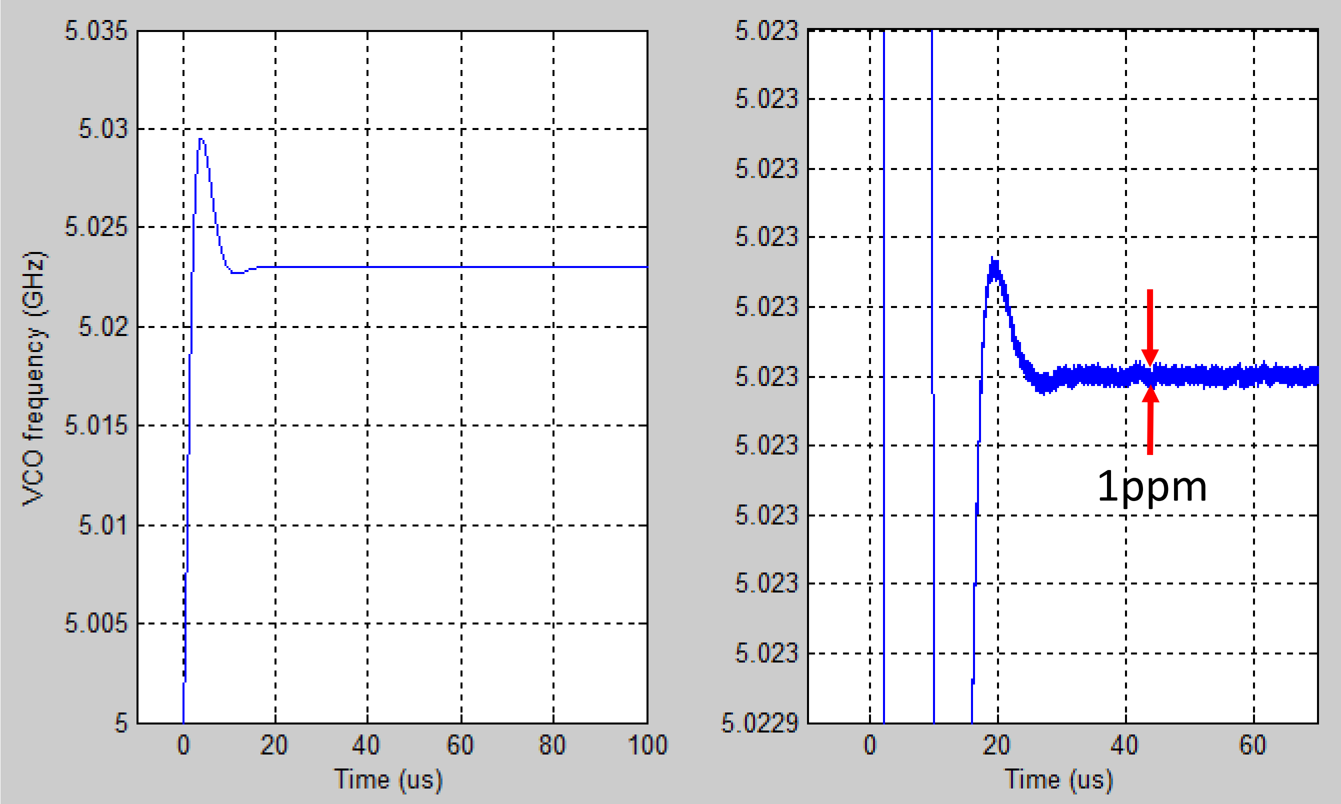
PLL design specifications: *Lock time*



- ❑ PLL lock time is not part of FOM but also an important spec for many applications
- ❑ Specified by time from PLL is powered on to output frequency or phase settles to given accuracy (ppm for frequency or degree for phase)
- ❑ Lock time includes
  - ❑ VCO and PLL start up time
  - ❑ VCO coarse tune time
  - ❑ VCO fine tune settling time
- ❑ Factors impact fine tune settling time
  - ❑ Loop bandwidth
  - ❑ Loop dynamics (phase margin)
  - ❑ Initial frequency and phase error

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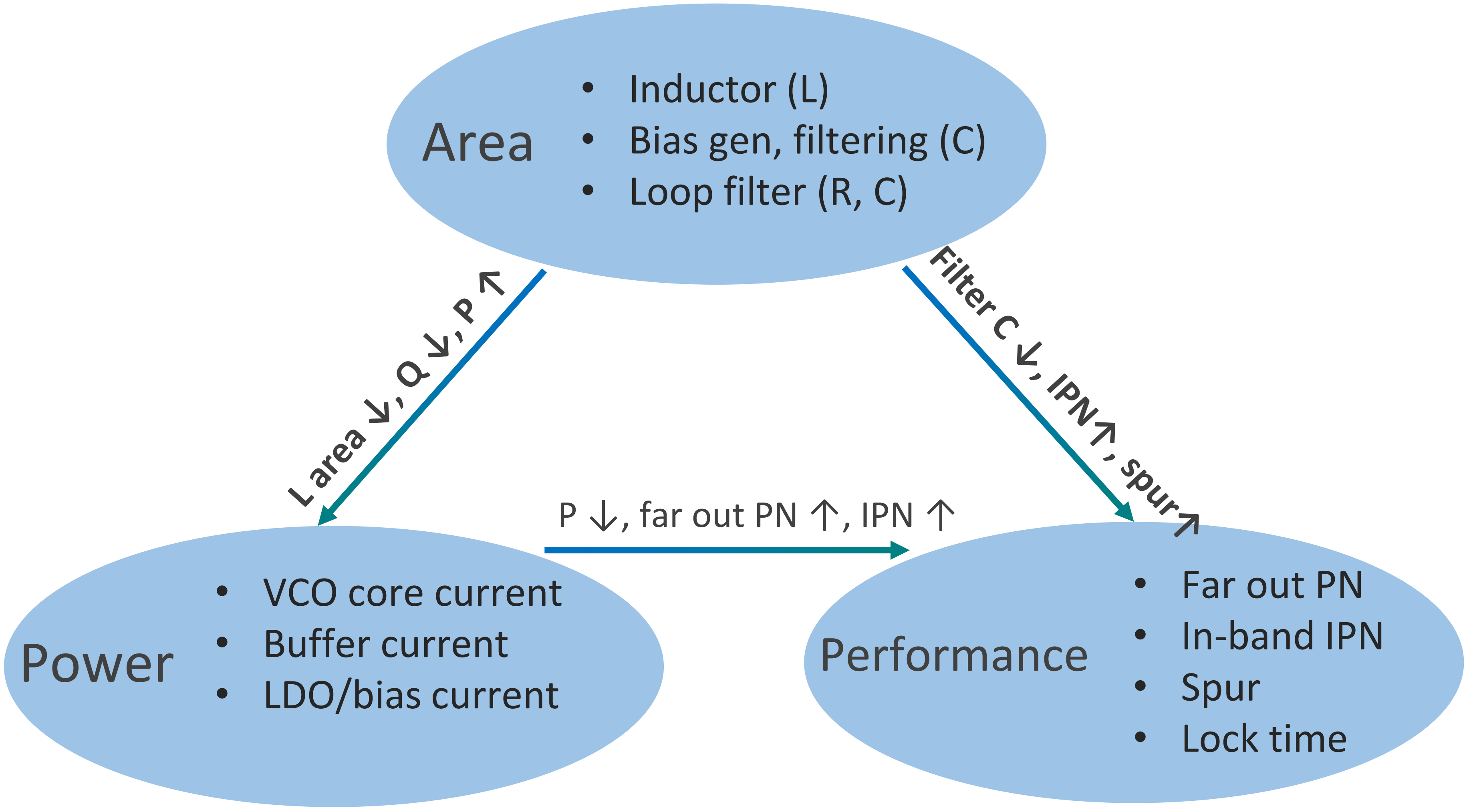
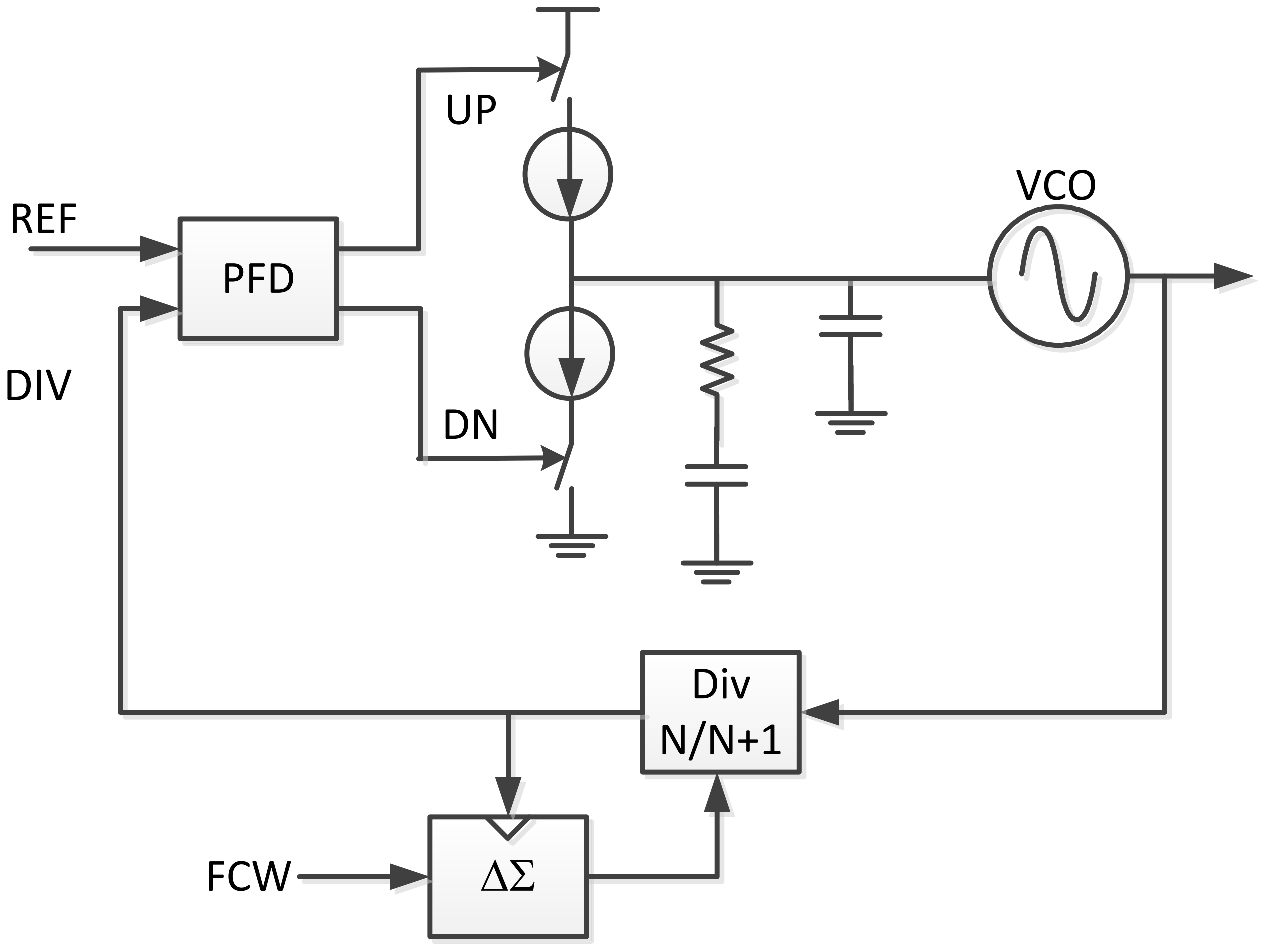
PLL design tradeoffs



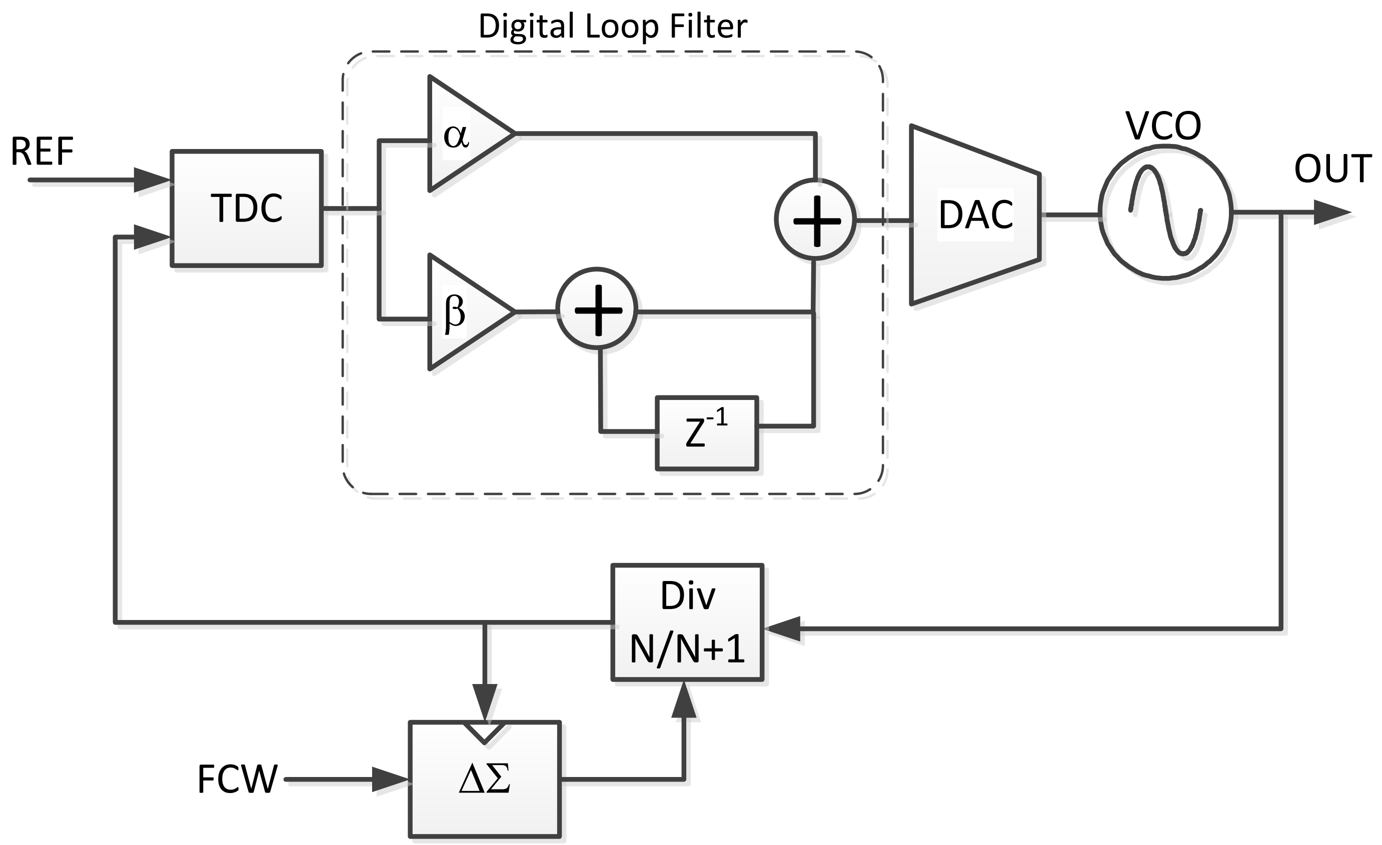
- Fundamental tradeoffs among area, power and performance
- Optimal topology to relax block level requirement to reduce area & power

Analog PLL



- Building blocks
  - Phase frequency detector (PFD)
  - Charge pump (CP)
  - Analog RC loop filter
  - Voltage controlled oscillator (VCO)
  - Frequency divider (NDIV)
  - Delta sigma modulator ($\Delta\Sigma$)
- With Input frequency f_{REF} and frequency control word (FCW) representing frequency multiplication ratio $N.\alpha$
  - Output frequency $F_{out} = N.\alpha * f_{REF}$

Digital PLL

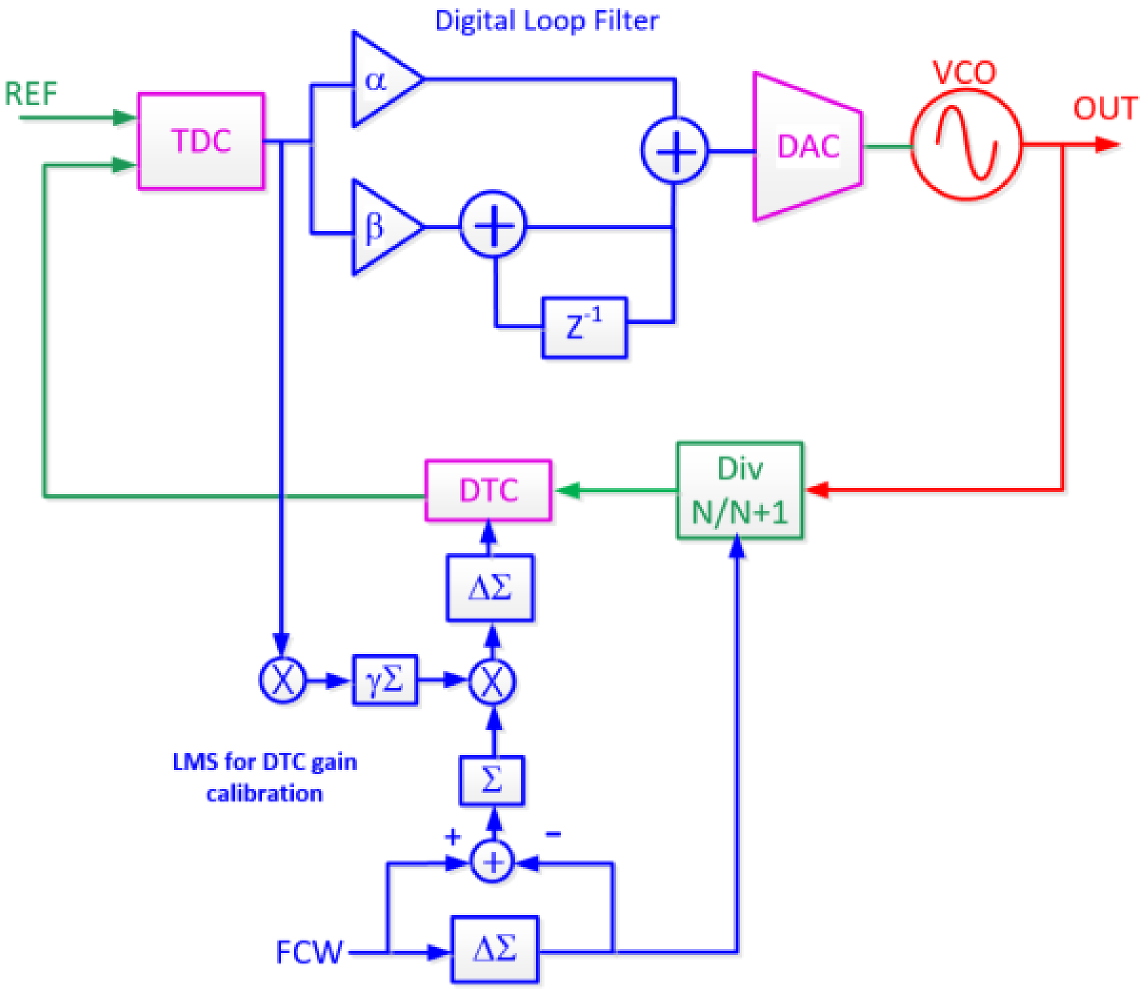


- Building blocks
  - Time to digital converter (TDC)
  - Digital loop filter
  - Digital to analog converter (DAC)
  - Voltage controlled oscillator (VCO)
  - Frequency divider (NDIV)
  - Delta sigma modulator (ΔΣ)
- With Input frequency f_{REF} and frequency control word (FCW) representing frequency multiplication ratio $N.\alpha$
  - Output frequency $F_{out} = N.\alpha * f_{REF}$

Analog PLL vs. digital PLL

	DPLL	APLL
Key Blocks	• TDC/PDC • Digital loop filter • NDIV • DCO/DAC+VCO	• PFD+CP • Sampling PD + Gm • Analog loop filter • NDIV • VCO
Pros ☺	• Compact digital loop filter • DSM noise cancellation • Simple TPM implementation	• Low IPN • Low digital spur
Cons ☹	• Quantization noise from TDC/PDC and DCO/DAC • Digital spur/noise to DCO	• Analog LF not tech scalable • DSM noise cancellation inaccurate

PLL Design Challenges



- ❑ VCO: RF or mmW frequency block
- ❑ Digital loop filter, ΔΣ modulator and LMS for DTC gain calibration: digital blocks
- ❑ TDC, DTC and DAC: analog and digital mixed signal blocks
- ❑ Frequency divider (prescaler, A/B counter): analog blocks

PLL design requires a broad range of skillset:

- RF/mmW design and EM modeling
- Analog and mixed signal block design
- Digital RTL, synthesis, place & route, timing closure
- System level simulation for spec definition
- RTL and AMS simulation for functionality verification

They make synthesizer design both challenging and interesting!

References

[1] Behzad Razavi, RF Microelectronics, Prentice Hall, 2nd edition, 2012

[2] William F. Egan, Frequency Synthesis by Phase Lock, 2nd edition, 2000

[3] N. Markulic, *et. al*., A self-calibrated 10Mb/s phase modulator with -37.4dBc EVM based on a 10.1-to-12.4GHz -246.6dB-FOM, fractional-N subsampling PLL, pp 176-177, ISSCC 2016

[4] D. Murphy, H. Darabi, A. Abidi, A. Hafez, A. Mirzaei, M. Mikhemar, F. Chang, A blocker-tolerant, noise-cancelling receiver suitable for wideband wireless applications, pp 294-314, JSSC 2012

[5] Z. Xu, M. Miyahara, K. Okada, A. Matsuzawa, A 3.6GHz low-noise fractional-N digital PLL using SAR-ADC-based TDC, pp 2345-2356, JSSC 2016

[6] C.-S. Chiu, *et. al*., A 40nm low-power transceiver for LTE-A carrier aggregation, ISSCC 2017.

[7] J. Lee, *et. al*., A Sub-6GHz 5G New Radio RF Transceiver Supporting EN-DC with 3.15Gb/s DL and 1.27Gb/s UL in 14nm FinFET CMOS, ISSCC 2019 & JSSCC Dec. 2019

[8] J. Deepthi, *et. al.*, A6.7–3.6-pJ/b 0.63–7.5-Gb/s rapid on/off clock and data recovery With <55ns turn-on time, IEEE Solid-State Letters, 2024

[9] M. Cusmai, *et. al.*, A 0.92-pJ/b PAM-4 and 0.61-pJ/b PAM-6 224-Gb/s DAC-Based Transmitter in 3-nm FinFET, JSSC 2025

[10] B. Zhang, *et. al.*, A 112-Gb/s Serial Link Transceiver With Three-Tap FFE and 18-Tap DFE Receiver for up to 43-dB Insertion Loss Channel in 7-nm FinFET Technology, JSSC 2024